

AI-Based Forest Cover Change Detection

North Eastern Region of India | Meghalaya & Nagaland

IIT Madras · Data Science & AI Project · Milestone 6

Team Members

Harsh kumar

Suranjan Das

Aditya Nupani

Shubham Mishra

Aman Kankriya

Project Overview

Problem

Detect pixel-level forest loss in Meghalaya & Nagaland using bi-temporal satellite imagery

Approach

Deep learning (Siamese U-Net + ASPP) fusing Sentinel-1 SAR and Sentinel-2 optical data

Challenge

Cloud cover, class imbalance (only 1.2% change pixels), small-scale Jhum cultivation

Outcome

Test F1 = 0.5348 · IoU = 0.3650 · ROC-AUC = 0.9776 · Deployed web application

Study Area & Context

Meghalaya & Nagaland — biodiversity hotspots of India's North East facing deforestation from Jhum (shifting) cultivation, infrastructure expansion, and urbanisation.

2021–2023 temporal window · 10 m resolution (Sentinel) · Custom dataset of 7,500 bi-temporal patches · 70/15/15 train-val-test split

Problem & Motivation

Why This Problem?

Traditional monitoring limitations:

- Hansen GFC: 30 m resolution — misses small-scale Jhum clearances
- Optical imagery: cloud cover disrupts data in monsoon (Jun–Sep)
- Manual methods: slow and non-scalable

Our Solution:

- SAR (Sentinel-1) penetrates cloud cover
- 10 m multi-modal fusion (SAR + optical)
- Deep learning for automated pixel-level detection

Cloud Cover

SAR solves it — cloud-penetrating microwave sensing

Class Imbalance

Only ~1.2% of pixels show forest loss; Focal + Dice loss used

Small Patches

Jhum clearances span just a few pixels at 30 m resolution

Label Noise

30 m Hansen labels resampled to 10 m introduce boundary error

Dataset Preparation Pipeline

1

Data Sources

Sentinel-1 SAR
(VV, VH)
Sentinel-2 Optical
(B2,B3,B4,B8)
Hansen GFC labels

2

GEE Processing

Thermal noise
removal · Radiometric
calibration · SRTM
terrain correction
3-month composites

3

Patch Generation

256×256 patches
6-channel input
(SAR+optical)
Label masks at
10 m resolution

4

Quality Checks

Cloud removal
Alignment verify
Abnormal pixel
filter · Non-forest
exclusion

5

Train/Val/Test

70% / 15% / 15%
Stratified by region
& change intensity
7,500 total samples

7,500

Bi-temporal
Patches

256×256

Patch Size
(pixels)

6 Ch

Input Channels
(SAR + Optical)

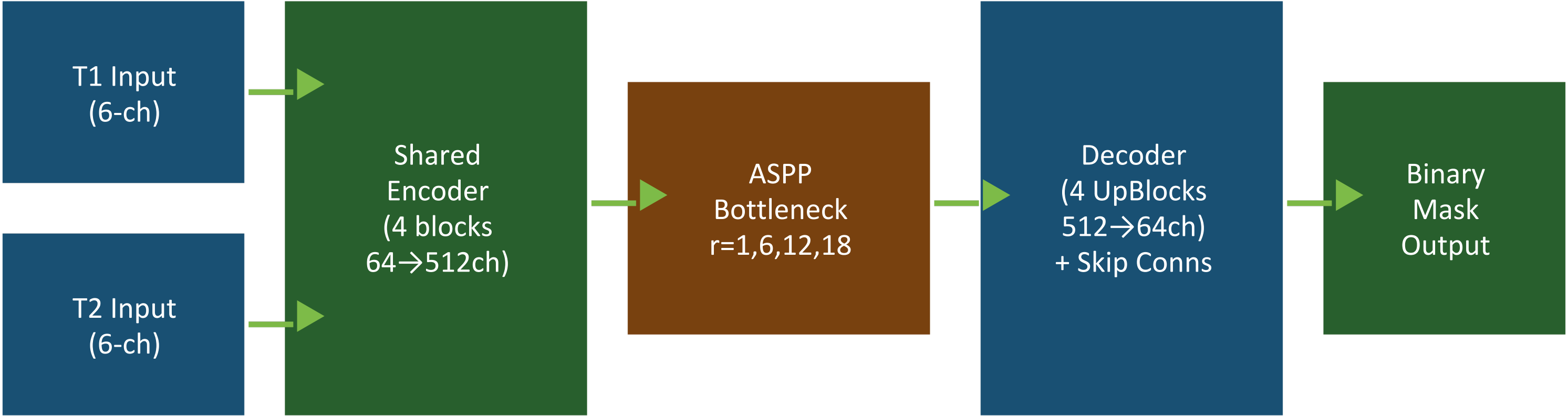
10 m

Spatial
Resolution

2021–23

Temporal
Window

Model Architecture: Siamese U-Net + ASPP



Key Design Choices

Siamese
Shared Weights

Feature
Differencing

ASPP
Multi-scale

Skip
Connections

Sigmoid
Output

Architecture Summary

Stage	Details	Channels
Encoder (4 blocks)	DoubleConv + MaxPool	6 → 512
ASPP Bottleneck	Atrous $r=\{1,6,12,18\}$ + Projection	512 → 512
Decoder (4 blocks)	TransposeConv + DoubleConv + Skip	512 → 64
Output	Conv1×1 + Sigmoid	64 → 1

Methodology: Loss Functions, Training & Enhancements

Loss Functions

Dice Loss (0.7×)

Optimises pixel overlap — robust to class imbalance

Focal Loss (0.3×)

Down-weights easy negatives; $\alpha=0.8$, $\gamma=2$

Combined L

$0.7 \cdot L_{\text{Dice}} + 0.3 \cdot L_{\text{Focal}}$ — dominant Dice for recall

Training Setup

Optimizer: Adam ($\text{lr}=1\text{e-}4$, $\beta_1=0.9$, $\beta_2=0.999$)

Batch size: 16 · Max epochs: 50 · Early stopping: patience 5

Scheduler: ReduceLROnPlateau (factor=0.5, patience=2)

Hardware: NVIDIA Tesla T4 (Kaggle) · AMP Mixed Precision

Tracking: Weights & Biases (W&B)

Experiments (Controlled Study)

E1

Baseline

Siamese U-Net + ASPP · BCE+Dice loss · from scratch

E2

Enhanced Training

Dice+Focal · data augmentation · dynamic threshold · LR scheduler

E3

ResNet-34 Encoder

Pretrained ImageNet encoder · 6-ch adaptation · domain gap challenge

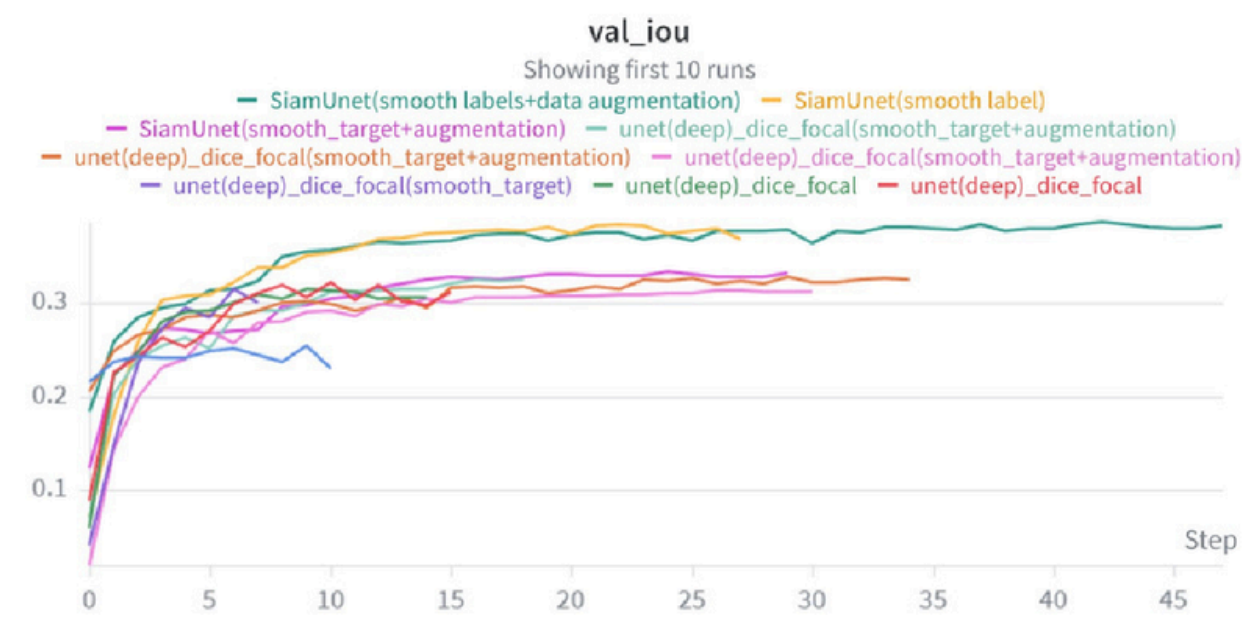
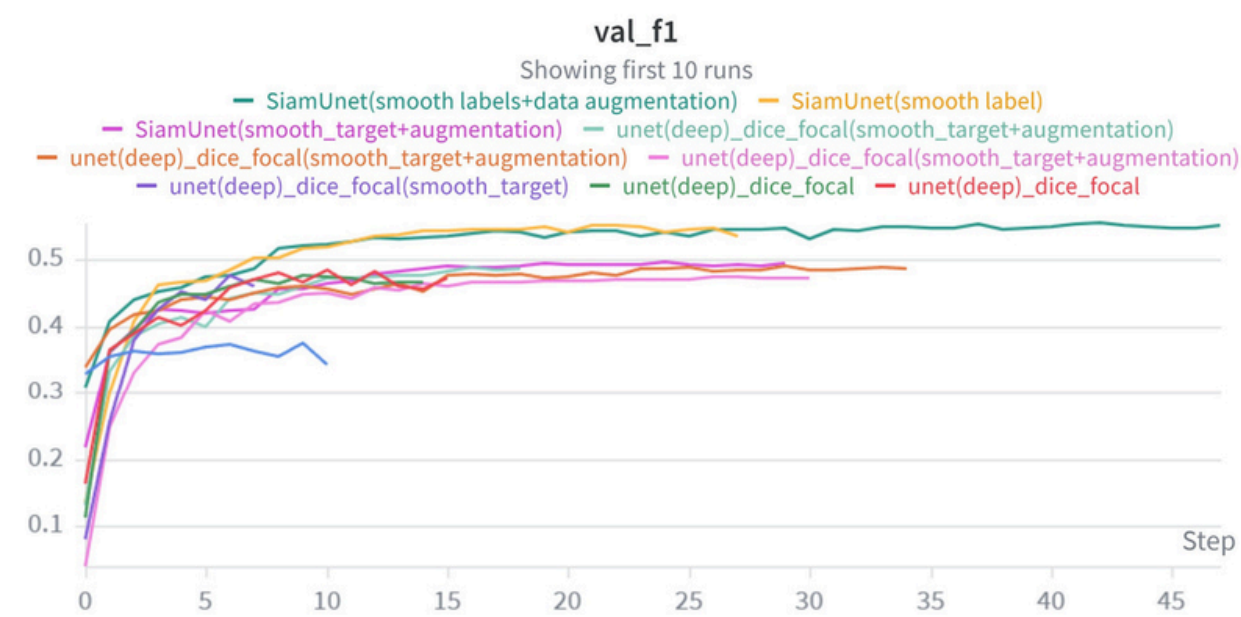
E4

SiameseUNet V2

Attention gates · ChangeFusion module · deep supervision

Experimental Results: Model Comparison

Model	Val F1	Val IoU	Best Epoch	Note
Baseline (BCE+Dice)	0.5060	0.3413	14	Reference model
Enhanced (Dice+Focal + Aug.) ★	0.5189	0.3526	23	Best model
ResNet-34 Encoder	0.4625	0.3026	12	Domain gap issue
SiameseUNet V2 (Attn+CF+DS)	0.4869	0.3240	12	Needs more data



Key Insight

Training-level improvements

(loss design + augmentation) outperform complex architectures at this dataset scale.

More data > More architecture

7,500 samples insufficient to leverage ResNet/attention benefits.

Test Set Evaluation (867 held-out samples)

0.5348

F1-Score

0.3650

IoU

0.9776

ROC-AUC

0.5141

PR-AUC

60.5%

Recall

99.2%

Specificity

Per-Category Performance — Enhanced SiameseUNet_ASPP | Threshold=0.3

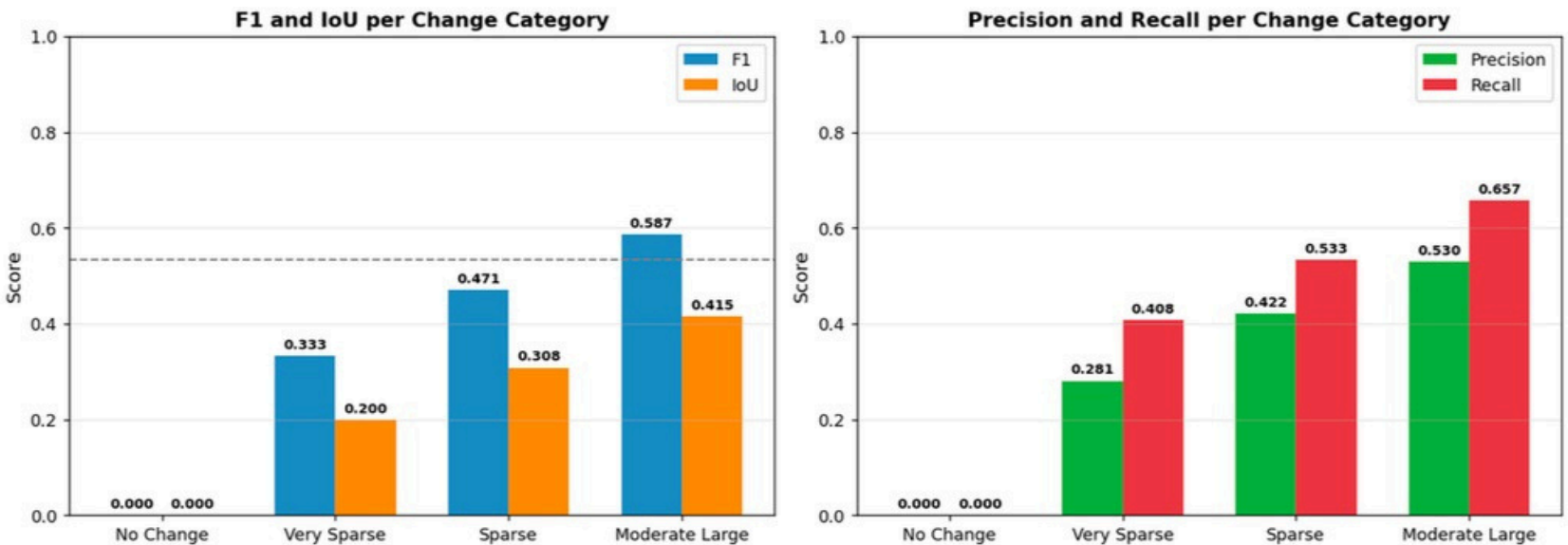


Figure 9: Per-category F1, IoU, Precision and Recall on the test set. Performance scales consistently with change intensity, with *moderate large* patches achieving the highest scores across all metrics.

Category	Samples	Avg GT px	Prec.	Recall	F1	IoU
No Change	97	0	0.000	0.000	0.000	0.000
Very Sparse	308	129	0.281	0.408	0.333	0.200
Sparse	301	690	0.422	0.534	0.471	0.308
Moderate-Large	161	2693	0.530	0.657	0.587	0.415
Overall	867	—	0.479	0.605	0.535	0.365

Table 7: Per-category pooled pixel-level metrics on the test set at threshold = 0.30. Avg GT px: average number of ground truth changed pixels per patch in that category.

Result Highlights

- **Threshold $\tau^* = 0.30$** (below 0.5 due to class imbalance)
- **ROC-AUC 0.9776** shows strong discriminative power
- **PR-AUC gap** (0.51 vs 0.98) reveals imbalance challenge
- **Large patches** detected well (F1=0.587)
- **Very sparse patches** hardest (F1=0.333)
- **No overfitting**: test metrics $\approx$ validation metrics

Error Analysis: Failure Modes

01 Scale Sensitivity

Performance drops sharply for small patches. Jhum clearances < 50 px (0.5 ha) at 10 m are missed — insufficient spatial context for the U-Net receptive field.

02 Systematic Under-Detection

Model assigns low confidence to genuine forest loss pixels. 269K false negatives vs 448K false positives. Label noise from 30m $\rightarrow$ 10m resampling ambiguates boundaries.

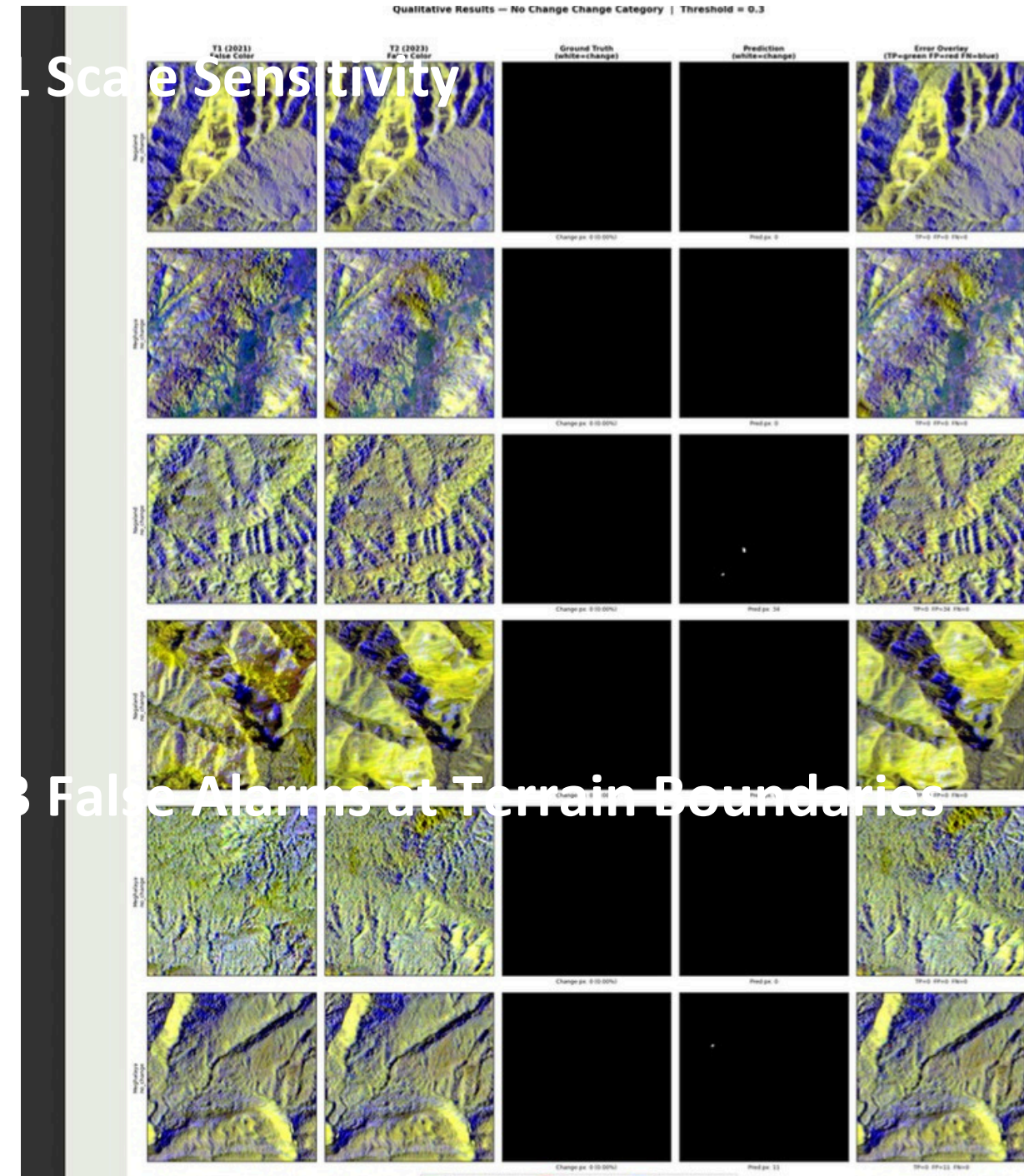
03 False Alarms at Terrain Boundaries

Ridgelines, river banks, and steep slopes mimic deforestation spectral signatures in NIR/Red bands. No DEM input means the model cannot distinguish topographic bare ground.

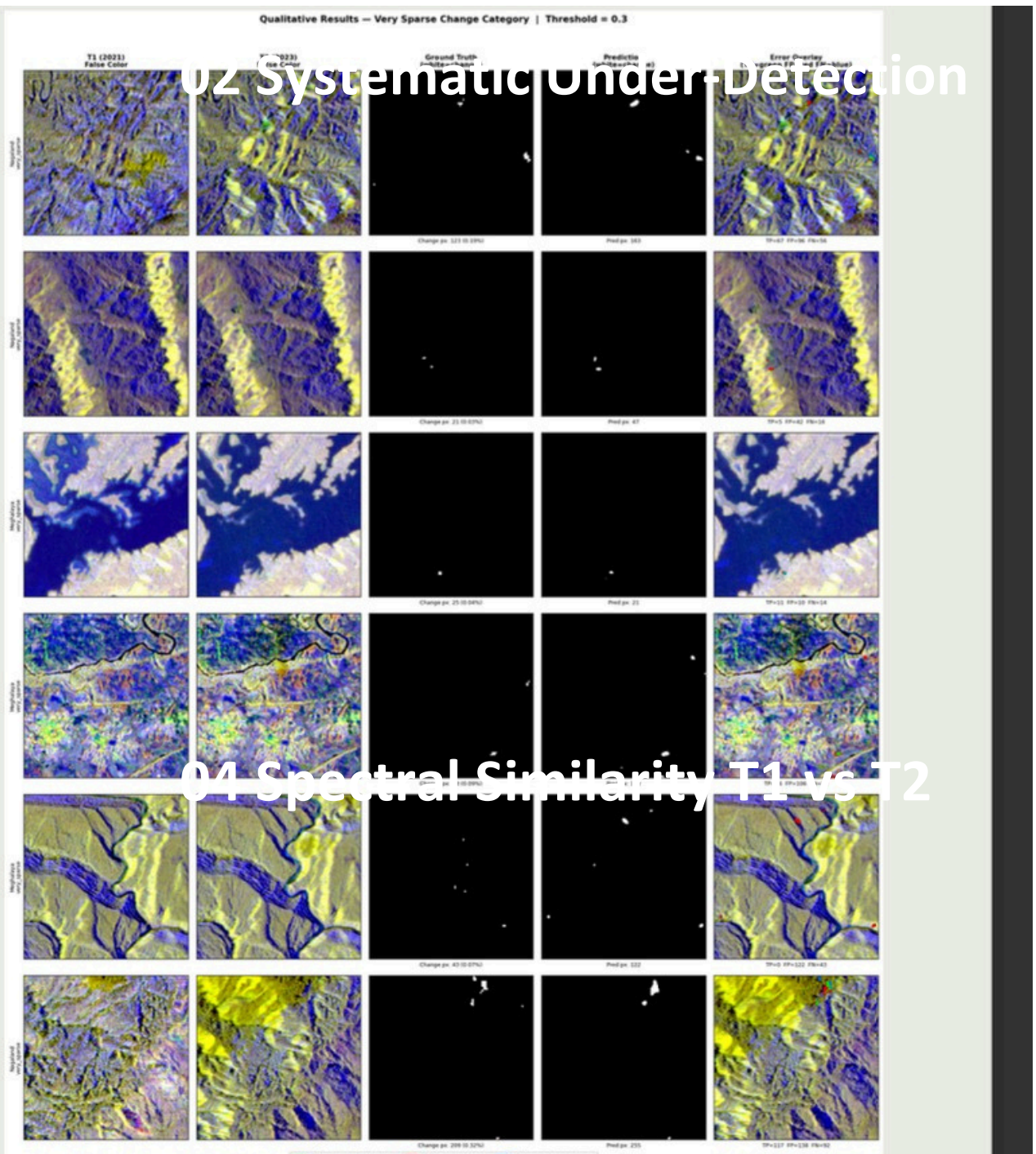
04 Spectral Similarity T1 vs T2

Jhum regrowth between acquisition dates makes T1 and T2 nearly identical despite confirmed loss in Hansen labels. Seasonal compositing may suppress genuine spectral contrasts.

Qualitative Results- Output Visualization



No Change Change Category



Very Sparse Change Category

Qualitative Results- Output Visualization

01 Scale Sensitivity

02 Systematic Under-Detection

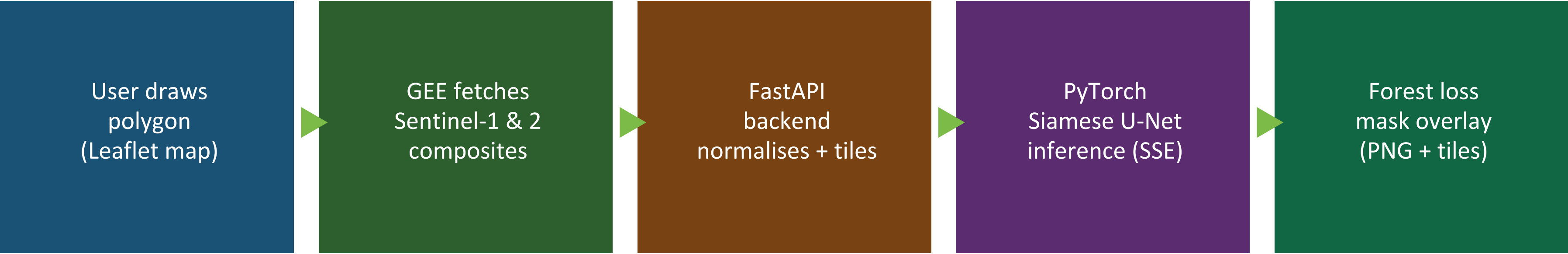
03 False Alarms at Terrain Boundaries

04 Spectral Similarity T1 vs T2

Sparse Change Category

Moderage Large
Change Category

Model Deployment & Real-Time Inference



Technology Stack

Component	Technology
Backend	FastAPI + Uvicorn (ASGI)
Satellite Data	Google Earth Engine Python API
Model Weights	Hugging Face Hub
Inference	PyTorch (CPU/CUDA auto-detect)
Streaming	Server-Sent Events (SSE)
Frontend Map	Leaflet.js
Containers	Docker
Hosting	Hugging Face Spaces

Interactive Features

- Draw any polygon on the map
- Select T1 and T2 years
- Live progress via streaming
- Adjustable decision threshold slider
- Results overlaid as transparent PNG

Live App: huggingface.co/spaces/suranjan90/forest-cover-loss

Limitations & Future Work

Current Limitations

- ✗ Only 7,500 patches — insufficient for complex architectures
- ✗ Label noise from 30m→10m Hansen resampling
- ✗ No cloud masking on Sentinel-2 composites
- ✗ Two-date only — cannot distinguish permanent vs seasonal change
- ✗ No DEM input — terrain features cause false alarms
- ✗ F1 target of 0.80 not yet reached (current: 0.53)

Future Directions

→ Dataset Expansion

More states, regions, and temporal windows

→ Multi-date Time Series

3–5 acquisition epochs to resolve Jhum regrowth ambiguity

→ Semi-supervised Learning

Exploit unlabelled satellite imagery for pretraining

→ Change Detection Transformers

BIT (Bitemporal Image Transformer) designed for SAR/optical

→ DEM Integration

Eliminate terrain-boundary false alarms with elevation data

→ Improved Label Quality

Higher-res reference data or active learning annotation

Conclusion & Key Takeaways

01 Working System Built

End-to-end forest change detection pipeline from satellite data acquisition to deployed web app — covering SAR + optical fusion, patch dataset, model training, and inference API.

02 Best Model: Enhanced Training

Dice+Focal loss + data augmentation + dynamic threshold achieves $F1=0.5348$, $IoU=0.3650$, $ROC-AUC=0.9776$ on the test set. Architecture < Training quality at 7,500 samples.

03 Scale Matters More

Pretrained ResNet-34 and attention-augmented V2 both underperformed the simple baseline — highlighting that dataset scale limits architectural complexity gains.

04 Deployed & Accessible

Interactive web app allows any user to draw a region, fetch live satellite imagery, and visualise predicted forest loss with an adjustable threshold — in real time.

Thank you